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39



BC210418563: SHAHZADI CHASHMAN

MTH401 - Differential Equations (Quiz#2)

Quiz Start T

Question # 10 of 10 (Start time: 08:17:52 PM, 24 December 2022)

If

$$y = c_1 e^{(-2+\sqrt{6})x} + c_2 e^{(-2-\sqrt{6})x}$$

is the complementary solution of

$$y'' + 4y' - 2y = 2x^2 - 3x + 6$$

,then the general form of its particular solution is

$$y_p$$

=_____.

Select the correct option



$$Ax^3 + Bx^2 + Cx + D$$



$$(Ax^2 + Bx)2x^2$$



$$Ax + B$$



$$Ax^2 + Bx + C$$

Click to Save Ar

8:16



4G



58%



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The hook's law states that the force F is proportional to the _____.

Select the correct option



Weight



Elongation



None of these



Length

Click to Save

8:15



4G 58%



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MTH401 - Differential Equations (Quiz#2)

Question # 6 of 10 (**Start time: 08:15:29 PM, 24 December 2022**)

The periodic time is given by

Select the correct option



$$\frac{\pi}{\omega}$$



$$2\pi * \omega$$



$$\frac{\omega}{2\pi}$$



$$\frac{2\pi}{\omega}$$

8:05



4G 59%



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39



MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 07:57 PM, 24 Decem

Question # 9 of 10 (Start time: 08:05:09 PM, 24 December 2022)

Total

If we want to solve an initial value problem then we need to to apply the initial conditions to the _____ solution and not to the complementary function alone.

Select the correct option



exact



general



complementary



particular

Click to Save Answer & Move to Next Que

BC200201330: IMRANA

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 07:02 PM, 24 December 20

Time Left 84 sec(s)

Question # 10 of 10 (Start time: 07:08:49 PM, 24 December 2022)

Total Mark

The root(s) of the auxiliary equation, $am^2 + bm + c = 0$, for the differential equation, $ay'' + by' + cy = 0$ is (are) _____.

Reload Math Equation

Select the correct option

☒
☐
☐
☐

(IV) Both (I) and (II).

(II) $m_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$

(III) $m_3 = \frac{b + \sqrt{b^2 + 4ac}}{2a}$

(I) $m_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$

Click to Save Answer & Move to Next Question

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BC200201330: IMRANA

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 07:02 P

Question # 9 of 10 (Start time: 07:08:25 PM, 24 December 2022)

An annihilator operator L for the function, $f(x) = e^{3x}$, is $L =$ _____.

Select the correct option

- | | |
|----------------------------------|--------------|
| ☐ | $3e^{3x}$ |
| ☐ | $D - e^{3x}$ |
| ☐ | e^{3x} |
| ☒ | $D - 3$ |

Reload

Click to Save Answer & Move to Next

200201330: IMRANA

H401 - Differential Equations (Quiz#2)

Quiz Start Time: 07:02 PM, 24 December 2022

Total Marks:

Question # 8 of 10 (Start time: 07:07:41 PM, 24 December 2022)

Following is (are) the form(s) of the general solution of the differential equation, $ay'' + by' + cy = 0$, corresponding to the case(s): _____.

Select the correct option

Reload Math Equations

(II) m_1 and m_2 real and equal i.e. $(b^2 - 4ac = 0)$

(I) m_1 and m_2 real and distinct i.e. $(b^2 - 4ac > 0)$

(IV) All (I), (II) and (III).

(III) m_1 and m_2 conjugate complex numbers i.e. $(b^2 - 4ac < 0)$

Click to Save Answer & Move to Next Question

BC200201330: IMRANA

Time Left

73

900

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 07:02 PM, 24 December 2022

Question # 7 of 10 (Start time: 07:06:39 PM, 24 December 2022)

Total Marks

The variation of parameters consists of finding a function $u_1(x)$ such that, $y_p = u_1(x) + y_1(x)$, is a _____ solution of the non-homogeneous differential equation, $\frac{dy}{dx} + P(x)y = f(x)$.

Select the correct option

Reload Math Equations

☐	complementary
☐	exact
☒	particular
☐	general

Click to Save Answer & Move to Next Question

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BC200201330: IMRANA

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 07:0

Question # 6 of 10 (Start time: 07:06:04 PM, 24 December 2022)

The solution of the auxiliary equation determines the _____ of the differential equation.

Select the correct option

☐	degree
☐	order
☐	linearity
☒	solutions

Click to Save Answer & M

MTH401 - Differential Equations (Quiz#2)

Question # 5 of 10 (Start time: 07:05:20 PM, 24 December 2022)

What is annihilator operator of the function

$$g(x) = 5$$

?

Select the correct option

☐ 5D☐ D+1☐ D+5☒ D

Click to

BC200201330: IMRANA

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 07:02 PM, 24 December 2022

Time Left: 64 sec(s)

Total Marks

Question # 4 of 10 (Start time: 07:04:34 PM, 24 December 2022)

Consider the non-homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = 5x + 7$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

☒

$Ax + B$

☐

$Ax^3 + Bx^2 + Cx + D$

☐

$Ax^2 + Bx + C$

☐

A

Click to Save Answer & Move to Next Question

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Question # 3 of 10 (Start time: 07:03:43 PM, 24 December 2022)

If

$$y = e^{5x}$$

, then which of the following is true for it?

Select the correct option

- | | |
|----------------------------------|---------------------------------------|
| ☐ | Its annihilator operator is $(D+5)$ |
| ☒ | Its annihilator operator is $(D-5)$ |
| ☐ | Its annihilator operator is $(D-5)^2$ |
| ☐ | Its annihilator operator is $(D+5)^2$ |

[Click to Save Answer](#)

BC200201330: IMRANA

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 07:02 PM,

Question # 2 of 10 (Start time: 07:03:23 PM, 24 December 2022)

The auxiliary equation of the differential equation, $y''' + 2y'' - 3y = 0$, is _____

Select the correct option

Reload M

- | | |
|----------------------------------|------------------------------|
| ☐ | $m^3 - 3 = 0$ |
| ☐ | $(m^3 + 2m^2 - 3)e^{mx} = 0$ |
| ☒ | $m^3 + 2m^2 - 3 = 0$ |
| ☐ | $m^3 + 2m^2 = 0$ |

Click to Save Answer & Move to Next Q

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Time Left

4

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 07:02 PM, 24 December 2022

Question # 1 of 10 (Start time: 07:02:13 PM, 24 December 2022)

Total Marks

For a second order linear ODE, if y_1 and y_2 are its solutions, and further $W(y_1, y_2) = 4$, $W_2 = -\frac{1}{5}$,
then $u'_2 =$ _____.

Select the correct option

 Reload Math Equation

- | | |
|----------------------------------|-----------------|
| ☐ | $-\frac{1}{5}$ |
| ☐ | -20 |
| ☐ | 4 |
| ☒ | $-\frac{1}{20}$ |

Click to Save Answer & Move to Next Question

F11



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Question # 10 of 10 (start time: 06:44:31 PM, 24 December 2022)

Total Marks: 1

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = \sin 4x$, then the assumed particular solution y_p could be _____.

Select the correct option

 Reload Math Equations

☐	$(A + B)(\cos 4x + \sin 4x)$
☐	Ae^{4x}
☐	$A \cos 4x + B \sin 4x$
☐	$Ax^2 + Bx + C$

Click to Save Answer & Move to Next Question

Time Left
sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 05:23 PM, 24 December 2022

Question # 9 of 10 (Start time: 06:43:11 PM, 24 December 2022)

Total Marks: 1

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = 3x^2 - 2$, then the assumed particular solution y_p could be _____.

Select the correct option

 Reload Math Equations

☐	$Ax + B$
☐	$Ax^3 + Bx^2 + Cx + D$
☐	$Ax^2 + Bx + C$
☐	A

Click to Save Answer & Move to Next Question

Question # 8 of 10 (Start time: 06:42:31 PM, 24 December 2022)

Total Marks: 1

If we want to solve an initial value problem then we need to to apply the initial conditions to the _____ solution and not to the complementary function alone.

Select the correct option

☐	complementary
☐	exact
☐	general
☐	particular

Click to Save Answer & Move to Next Question

Question # 7 of 10 (Start time: 06:40:57 PM, 24 December 2022)

Total Marks: 1

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x)$.
 If the input function, $g(x) = 5x^2 \sin 4x$, then the assumed particular solution y_p could be _____.

Select the correct option

 Reload Math Equations

☐	$(Ax + B)e^{3x} \cos 4x + (Cx + D)e^{3x} \sin 4x$
☐	$(A_1 x^2 + B_1 x + C_1) \cos 4x + (A_2 x^2 + B_2 x + C_2) \sin 4x$
☐	$A \cos 4x + B \sin 4x$
☐	$Ae^{3x} \cos 4x + Be^{3x} \sin 4x$

Click to Save Answer & Move to Next Question

Question # 6 of 10 (start time: 06:40:07 PM, 24 December 2022) Total Marks: 1

If a force acts upon a body, the acceleration is produced in the direction of the force whose magnitude is proportional to the magnitude of force, is known as the _____.

Select the correct option

☐	Newton's third law
☐	Newton's second law
☐	Newton's first law
☐	Hook's law

Click to Save Answer & Move to Next Question

6:03



4G



Quiz

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MTH401 - Differential Equations (Quiz#2)

Question # 6 of 10 (**Start time: 06:03:15 PM, 24 December 2022**)Wronskian of the functions, $y_1(x) = 1$, $y_2(x) = \sin x$, $y_3(x) = \cos x$, is _____

Select the correct option



2



-1



0



1

Following is (are) the form(s) of the general solution of the differential equation, $ay'' + by' + cy = 0$, corresponding to the case(s): _____.

Select the correct option



Reload Math Equation

☐

(II) m_1 and m_2 real and equal i.e. $(b^2 - 4ac = 0)$

☐

(I) m_1 and m_2 real and distinct i.e. $(b^2 - 4ac > 0)$

☐

(III) m_1 and m_2 conjugate complex numbers i.e. $(b^2 - 4ac < 0)$

☐

(IV) All (I), (II) and (III).

Click to Save Answer & Move to Next Question

Question # 10 of 10 (Start time: 05:08:26 PM, 24 December 2022)

Total Marks:

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = 3x^2 - 2$, then the assumed particular solution y_p could be _____.

Select the correct option

 Reload Math Equations

- | | |
|-----------------------|------------------------|
| ☐ | A |
| ☐ | $Ax^3 + Bx^2 + Cx + D$ |
| ☐ | $Ax^2 + Bx + C$ |

Start



5:09 PM
12/24/2022

Question # 9 of 10 (Start time: 05:07:45 PM, 24 December 2022)

Total Marks

If

$$y = 2 + x$$

, then which of the following is true for it?

Select the correct option

 Reload Math Equations

☐

Its annihilator operator is D

☐

Its annihilator operator is

$$D^2$$

☐

Its annihilator operator is

$$D^3$$

Start



5:08 PM
12/24/2022

Question # 8 of 10 (Start time: 05:06:20 PM, 24 December 2022)

Total Marks:

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = x e^{3x} \cos 4x$, then the assumed particular solution y_p could be _____.

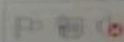
Select the correct option

 Reload Math Equations

☐ $A e^{3x} \cos 4x + B e^{3x} \sin 4x$

☐ $A \cos 4x + B \sin 4x$

Start

5:06 PM
12/24/2022

If the auxiliary equation, $am^2 + bm + c = 0$, has the complex roots, $m_1 = \alpha + i\beta$ and $m_2 = \alpha - i\beta$ where α and $\beta > 0$ are real and $i^2 = -1$. Then the solution of the differential equation, $ay'' + by' + cy = 0$ could be _____.

Select the correct option

 Reload Math Equation

$$y = c_1 e^{(\alpha + i\beta)x} + c_2 e^{(\alpha - i\beta)x}$$



$$y = c_1 e^{(\alpha + i\beta)x} - c_2 e^{(\alpha - i\beta)x}$$



$$y = c_1 e^{(\alpha + i\beta)x}$$

Start

5:05 PM
12/24/2022

Question # 5 of 10 (Start time: 05:03:23 PM, 24 December 2022)

Total Marks: 1

The variation of parameters consists of finding a function $u_1(x)$ such that, $y_p = u_1(x) + y_1(x)$, is a _____ solution of the non - homogeneous differential equation, $\frac{dy}{dx} + P(x)y = f(x)$.

Select the correct option



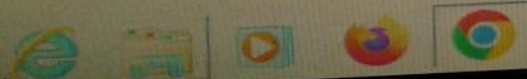
Reload Math Equations

particular

exact

general

rt



Select the correct option

☐	closure
☐	nonlinearity
☐	linearity
☐	inverse

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Google Chrome

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 04:59 PM, 24

Question # 4 of 10 (Start time: 05:01:57 PM, 24 December 2022)

The differential operator D possesses the _____ property.

Select the correct option

☐ closure

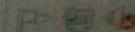
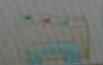


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MTH401 Differential Equations (Quiz#2)

Quiz Start Time: 04:59 PM, 24 Dec

Question # 3 of 10 (Start time: 05:01:27 PM, 24 December 2022)

The gravitational force exerted by the earth on a body of mass m is called ----- of the body.

Select the correct option

Force



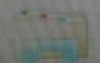
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MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 04:5

Question # 2 of 10 (Start time: 05:00:51 PM, 24 December 2022)

The periodic time is given by

Select the correct option

☐

$$\frac{\pi}{\omega}$$



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Differential Equations (Quiz#2)

Quiz Start Time: 04:59 PM, 24 Dec

of 10 (Start time: 04:59:39 PM, 24 December 2022)

The annihilator operator for the function, $g(x) = 2e^x$, is _____.

correct option

 Reload Math E

$$D + 1$$

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ed

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12

Question # 10 of 10 (Start time: 04:58:06 PM, 24 December 2022)

Total Marks: 1

The roots of the auxiliary equation, $m^2 - 10m + 25 = (m - 5)^2 = 0$, for a differential equation is _____.

Select the correct option

Reload Math Equations

- | | |
|-----------------------|------------------|
| ☐ | $m_1 = m_2 = 5$ |
| ☐ | $m = 25$ |
| ☐ | $m = -10$ |
| ☐ | $m_1 = m_2 = 25$ |

Question # 9 of 10 (Start time: 04:56:32 PM, 24 December 2022)

Total M

If we want to solve an initial value problem then we need to to apply the initial conditions to the _____ solution and not to the complementary function alone.

Select the correct option

☐ particular

☐ complementary

☐ general

☐ exact

← → ↻ vulms.vu.edu.pk/Quiz/QuizQuestion.aspx?ver=56cc4b52-804e-4a94-8223-398ff05749f0

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Question # 8 of 10 (Start time: 04:55:37 PM, 24 December 2022) Total Marks: 1

The velocity of a particle moving with simple harmonic motion is ----- at the mean position.

Select the correct option

☐	None of these
☐	Minimum
☐	Zero
☐	Maximum

Click to Save Answer & Move to Next Question

Type here to search 16°C Partly sunny 3:55 AM 12/24/2022

Question # 7 of 10 (Start time: 04:54:21 PM, 24 December 2022)

Total Marks: 1

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = 5x^2 \sin 4x$, then the assumed particular solution y_p could be _____.

Select the correct option

[Reload Math Equations](#)☐

$$Ae^{3x} \cos 4x + Be^{3x} \sin 4x$$

☐

$$(Ax + B)e^{3x} \cos 4x + (Cx + D)e^{3x} \sin 4x$$

☐

$$A \cos 4x + B \sin 4x$$

☐

$$(A_1 x^2 + B_1 x + C_1) \cos 4x + (A_2 x^2 + B_2 x + C_2) \sin 4x$$



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Question # 6 of 10 (Start time: 04:53:02 PM, 24 December 2022) Total Marks: 1

For a system in simple harmonic motion which of the following is the time required to complete a cycle of motion?

Select the correct option

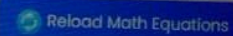
☐	Period
☐	Amplitude
☐	Revolution
☐	Frequency

[Click to Show Answer & Move to Next Question](#)

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If the auxiliary equation, $am^2 + bm + c = 0$, has repeated real roots with $m_1 = m_2$ then the solution of the differential equation, $ay'' + by' + cy = 0$, could be _____.

Select the correct option

 Reload Math Equations

☐ $y_2 = e^{m_1 x} + c e^{m_2 x}$

☐ $y = e^m$

☐ $y_2 = e^{m_1 x} + e^{m_2 x}$

☐ $y_1 = e^{m_1 x}$

The differential operator L is called an annihilator operator of the function f if: _____.

Select the correct option

- ☐ (iv) All (i), (ii) and (iii).
- ☐ (iii) The function f is such that $L(y)=0$.
- ☐ (i) L is a linear differential operator with constant coefficients.
- ☐ (ii) $y = f(x)$ defines a sufficiently differentiable function.

Click to Save Answer & Move to Next Question

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = 5x + 7$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

☐ $Ax^3 + Bx^2 + Cx + D$

☐ $Ax^2 + Bx + C$

☐ A

☐ $Ax + B$

BC190400449: LARAIB ARSHAD

MTH401 - Differential Equations (Quiz#2)

Time Left 72
sec(s)

Quiz Start Time: 04:48 PM, 24 December 2022

Question # 2 of 10 (Start time: 04:48:55 PM, 24 December 2022)

Total Marks: 1

If the auxiliary equation, $am^2 + bm + c = 0$, has the complex roots, $m_1 = \alpha + i\beta$ and $m_2 = \alpha - i\beta$, where α and $\beta > 0$ are real and $i^2 = -1$. Then the solution of the differential equation, $ay'' + by' + cy = 0$, could be _____.

Select the correct option

Reload Math Equations

☐

$$y = c_1 e^{(\alpha + i\beta)x} - c_2 e^{(\alpha - i\beta)x}$$

☐

$$y = c_1 e^{(\alpha + i\beta)x} + c_2 e^{(\alpha - i\beta)x}$$

☐

$$y = c_1 e^{(\alpha + i\beta)x}$$

☐

$$y = c_1 e^{(\alpha - i\beta)x}$$

Type here to search

Great sunset 3:49 AM 12/24/2022

Question # 1 of 10 (Start time: 04:48:06 PM, 24 December 2022)

The annihilator operator for the function, $g(x) = 5x$, is _____.

Select the correct option

Reload Math Equ

☐

$D + 1$

☐

D^2

☐

D

☐

$D + 5$

MTH401 - Differential Equations (Quiz#2)

Total Questions :10

Please read the following instructions carefully!

1. Quiz will be based upon Multiple Choice Questions (MCQs).
2. You have to attempt the quiz online. You can start attempting the quiz any time within given date(s) of a particular subject by clicking the link for Quiz in VULMS.
3. Each question has a fixed time of 90 seconds. So you have to save your answer before 90 seconds. But due to unstable internet speeds, it is recommended that you should save your answer within 60 seconds. While attempting a question, keep an eye on the remaining time.
4. Attempting quiz is unidirectional. Once you move forward to the next question, you can not go back to the previous one. Therefore before moving to the next question, make sure that you have selected the best option.
5. **DO NOT** press **Back Button** / **Backspace Button** while attempting a question, otherwise you will lose that question.
6. **DO NOT** refresh the page unnecessarily, **specially** when following messages appear
Saving...
Question Timeout! Now loading next question...
7. **Javascript MUST be enabled** in your browser; otherwise you will not be able to attempt the quiz.
8. If for any reason, you lose access to internet (like power failure or disconnection of internet), you will be able to attempt the quiz again from the question next to the last shown question. But remember that you have to complete the quiz before expiry of the deadline.
9. If any student failed to attempt the quiz in given time then no re-take or offline quiz will be held.

Important Announcement

Type here to search

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For non-homogeneous differential equations, the form of the particular solution, $y_p = y_{p_1} + y_{p_2} + \dots + y_{p_m}$, is a linear combination of all the linearly _____ functions generated by repeated differentiation of the input function $g(x)$.

option

Reload Math

independent

dependent

Click to Save



Question # 8 of 10 (Start time: 04:28:58 PM, 24 December 2022)

If the auxiliary equation, $am^2 + bm + c = 0$, has repeated real roots with $m_1 = m_2$ then the solution of the differential equation, $ay'' + by' + cy = 0$, could be _____.

Select the correct option

Reload

☐

$$y_2 = e^{m_1 x} + e^{m_2 x}$$

☐

$$y = e^m$$

☐

$$y_2 = e^{m_1 x} + C e^{m_2 x}$$

☐

$$y_1 = e^{m_1 x}$$

Click to Save Answer & Move to Next Question

Consider the non-homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = e^{5x}$, then the assumed particular solution y_p could be _____.

Select option

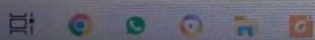
$$(Ax^2 + Bx + C)e^{5x}$$

$$Ae^{5x}$$

$$(Ax + B)e^{5x}$$

$$Ae^{5x} \cos 5x + Be^{5x} \sin 5x$$

Click to Save Answer & Move



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Question # 6 of 10 (Start time: 04:27:40 PM, 24 December 2022)

The annihilator operator for the function, $g(x) = 2e^{-x}$, is _____.

Select the correct option

- | | |
|-----------------------|---------|
| ☐ | D |
| ☐ | D^2 |
| ☐ | $D + 2$ |
| ☐ | $D + 1$ |

[Click to Show Answer](#)

Question # 5 of 10 (Start time: 04:26:42 PM, 24 December 2022)

If

is the complementary solution of

$$y = c_1 e^x + c_2 e^{4x}$$

, then the general form of its particular solution is

$$y'' - 5y' + 4y = 8e^x$$

y_p

Select the correct option

- ☐
- ☐
- ☐
- ☐

$$Ae^x$$

$$Axe^x$$

$$(Ax + B)e^{4x}$$

None of these

Search



13°C Smoke

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x)$.

If the input function, $g(x) = x e^{3x} \cos 4x$, then the assumed particular solution y_p could be _____.

correct option

Reload Math Equa

$$A e^{3x} \cos 4x + B e^{3x} \sin 4x$$

$$A \cos 4x + B \sin 4x$$

$$(A_1 x^2 + B_1 x + C_1) \cos 4x + (A_2 x^2 + B_2 x + C_2) \sin 4x$$

$$(Ax + B) e^{3x} \cos 4x + (Cx + D) e^{3x} \sin 4x$$

Click to Save Answer & Move to Next Question

$g(x)$

Annihilator

1

D

x

D^2

x^n

D^{n+1}

$e^{\alpha x}$

$(D - \alpha)$

$x e^{\alpha x}$

$(D - \alpha)^2$

$x^n e^{\alpha x}$

$(D - \alpha)^{n+1}$

$\sin(\beta x)$

$D^2 + \beta^2$

BC210421318: AREEBA QAMAR

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 04:20 PM, 24 December

Question # 4 of 10 (Start time: 04:23:18 PM, 24 December 2022)

$$\begin{aligned} & \text{Consider the non-homogeneous linear differential equation} \\ & y^{(n)} + a_{n-1}y^{(n-1)} + \dots + a_1y' + a_0y = g(x) \end{aligned}$$

If the input function $g(x) = x^3 \cos 4x$, then the assumed particular solution y_p could be $y_p = x^3 \cos 4x$.

Search



13°C

Smoke

4:24 PM

Question # 3 of 10 (Start time: 04:22:28 PM, 24 December 2022)

What is annihilator operator of the function

$$g(x) = 5$$

?

Select the correct option

- ☐ $D+5$
- ☐ $D+1$
- ☐ D
- ☐ $5D$

The root(s) of the auxiliary equation, $am^2 + bm + c = 0$, for the differential equation, $ay'' + by' + cy = 0$, is (are) _____.

correct option

(II) $m_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$

(III) $m_3 = \frac{b + \sqrt{b^2 + 4ac}}{2a}$

(I) $m_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$

(IV) Both (I) and (II).

Click to Save Answer

Question # 1 of 10 (Start time: 04:20:58 PM, 24 December 2022)

The polynomial linear differential operators with constant coefficients -----.

Select the correct option

☐ commute

☐ distributive

☐ identical

☐ inverse

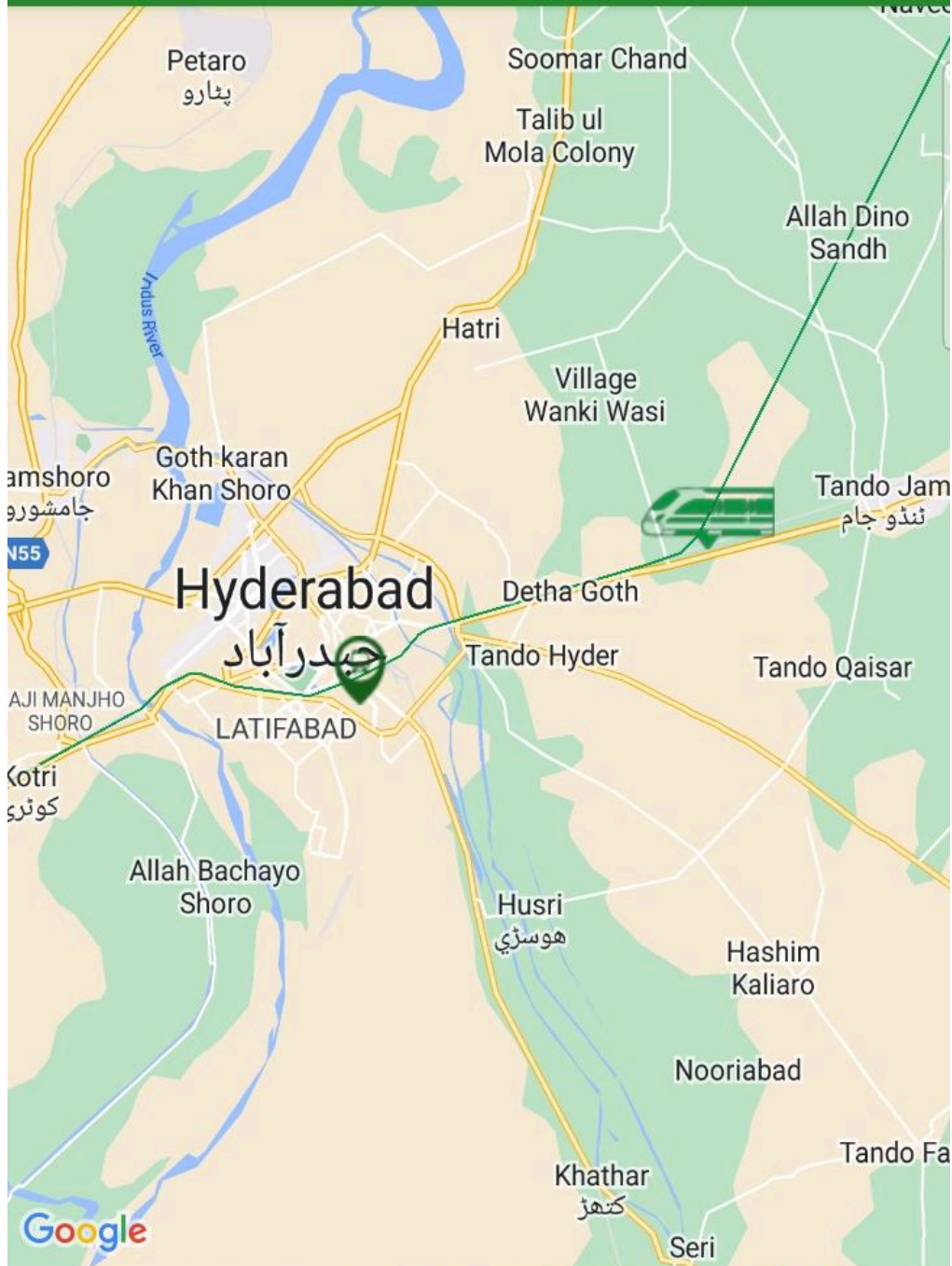
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	Departure Date	23 Dec
	Last Updated	just now
	Next Station	Hyderabad Jn
	Estimated Arrival	03:56 pm
	Delay	8hrs 1min
	Current Speed	49 Km/Hr

PRE ARRIVAL NOTIFICATION

Select the correct option

- ☐ (iv) All (i), (ii) and (iii).
- ☐ (iii) The function f is such that $L(y)=0$.
- ☐ (ii) $y = f(x)$ defines a sufficiently differentiable function.
- ☐ (i) L is a linear differential operator with constant coefficients.



Quiz - Google Chr...



To get future Google Chrome updates, you'll need Windows 10 or later. This computer is using Windows 7.

Question # 10 of 10 (Start time: 03:10:12 PM, 24 December 2022)

The differential operator L is called an annihilator operator of the function f if: _____

Select the correct option

- ☐ (iv) All (i), (ii) and (iii).
- ☐ (iii) The function f is such that $L(y)=0$.
- ☐ (ii) $y = f(x)$ defines a sufficiently differentiable function.
- ☐ (i) L is a linear differential operator with constant coefficients.

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Question # 9 of 10 (**Start time: 03:09:19 PM, 24 December 2022**)

In differential equations, the symbol D is known as _____ operator.

Select the correct option

- ☐ arithmetic
- ☐ relational
- ☐ differential
- ☐ integral



Quiz - Google Chr...



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ure Google Chrome updates, you'll need Windows 10 or later. This computer is using Windows 7.

n # 8 of 10 (**Start time: 03:08:38 PM, 24 December 2022**)

Total

Consider the non-homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = (9x - 2)e^{5x}$, then the assumed particular solution y_p could be _____.

the correct option

Reload Math Equatio

$$Ae^{5x} \cos 5x + Be^{5x} \sin 5x$$

$$(Ax^2 + Bx + C)e^{5x}$$

$$Ae^{5x}$$

$$(Ax + B)e^{5x}$$

Click to Save Answer & Move to Next Question



Quiz - Google Clt...



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3:08 PM
12/24/2022

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Question # 7 of 10 (**Start time: 03:08:01 PM, 24 December 2022**)

The periodic time is given by

Select the correct option

- | | |
|-----------------------|-----------------------|
| ☐ | $2\pi * \omega$ |
| ☐ | $\frac{\omega}{2\pi}$ |
| ☐ | $\frac{2\pi}{\omega}$ |
| ☐ | $\frac{\pi}{\omega}$ |

Click to Save A



Quiz - Google Chr...



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Question # 6 of 10 (Start time: 03:06:54 PM, 24 December 2022)

If

$$y = 2 + x$$

, then which of the following is true for it?

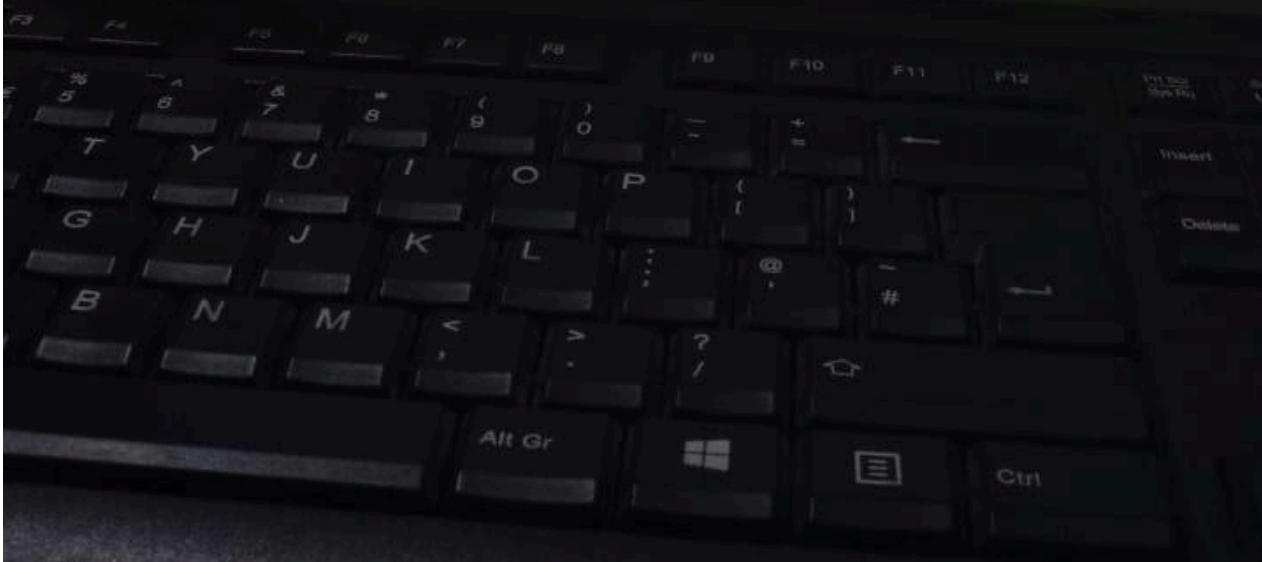
Select the correct option

- | | |
|-----------------------|-----------------------------------|
| ☐ | Its annihilator operator is D^3 |
| ☐ | Its annihilator operator is D |
| ☐ | Its annihilator operator is D^2 |
| ☐ | Its annihilator operator is $D+1$ |

Click to Save



Quiz - Google Chr...



Quiz

x

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Learn more X

Question # 5 of 10 (Start time: 03:05:57 PM, 24 December 2022)

Total Marks: 1

If the auxiliary equation, $am^2 + bm + c = 0$, has two unequal real roots m_1 and m_2 then the solutions of the differential equation, $ay'' + by' + cy = 0$, could be _____.

Select the correct option

Reload Math Equations

☐

(II) $y_1 = e^{m_1 x}$

☐

(I) $y = c e^{m x}$

☐

(IV) Both (II) and (III).

☐

(III) $y_2 = e^{m_2 x}$

Click to Save Answer & Move to Next Question



EN

3:06 PM
12/24/2022

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Question # 4 of 10 (**Start time: 03:05:26 PM, 24 December 2022**)

The polynomial linear differential operators with constant coefficients _____.

Select the correct option

☐

inverse

☐

commute

☐

distributive

identical



Quiz - Google Chr...



Quiz

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Question # 3 of 10 (Start time: 03:04:52 PM, 24 December 2022)

Total Marks

The roots of the auxiliary equation, $m^2 - 10m + 25 = (m - 5)^2 = 0$, for a differential equation is _____.

Select the correct option

Reload Math Equations

- | | |
|-----------------------|------------------|
| ☐ | $m_1 = m_2 = 25$ |
| ☐ | $m = 25$ |
| ☐ | $m = -10$ |
| ☐ | $m_1 = m_2 = 5$ |

Click to Save Answer & Move to Next Question

EN 3:05 PM
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Quiz

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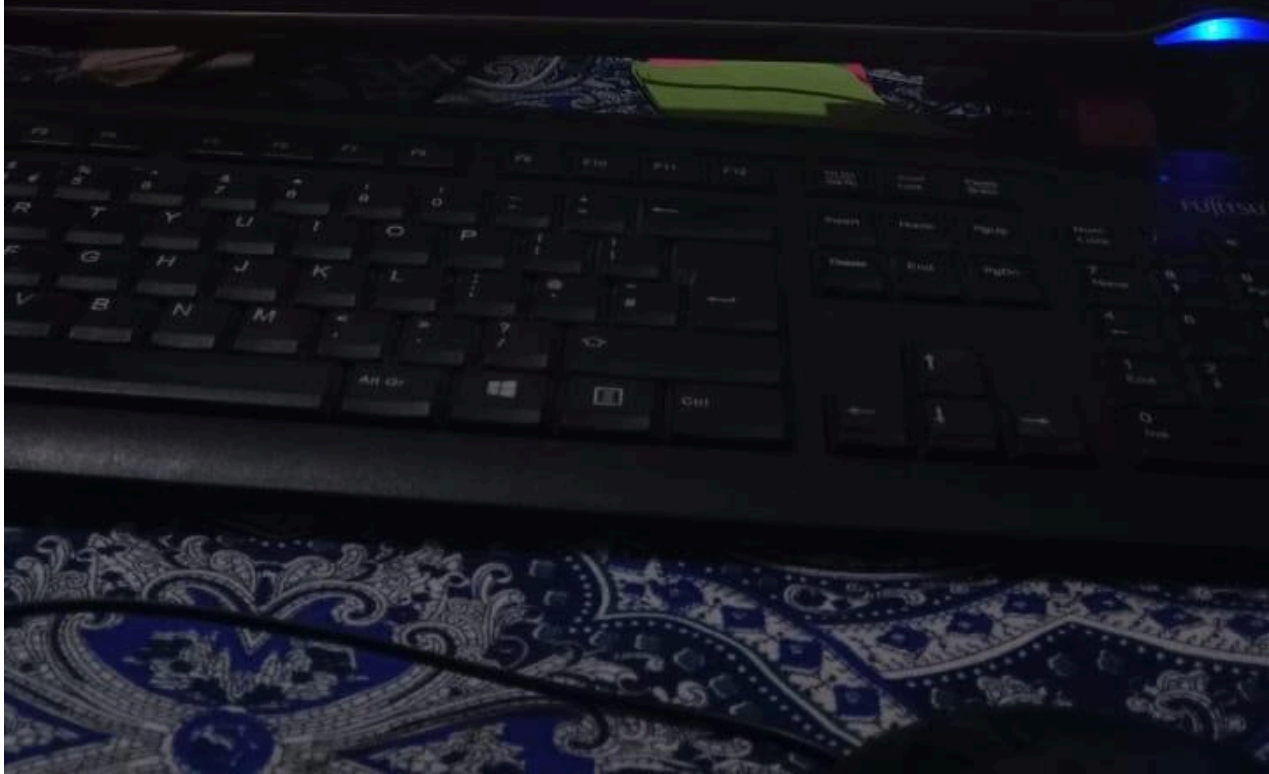
We can derive a differential equation governing the motion of a mass attached to spring when the Newton's second law combined with: _____

Select the correct option

☐	Newton's 3rd law
☐	Hook's law

Click to Save Answer & Move to Next Question

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Quiz

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Learn more

Question # 1 of 10 (Start time: 03:03:40 PM, 24 December 2022)

Total Mar

Consider the non-homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = 3x^2 - 2$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

☐

$Ax + B$

☐

A

☐

$Ax^2 + Bx + C$

☐

$Ax^3 + Bx^2 + Cx + D$

Click to Save Answer & Move to Next Question

EN

3:03 PM
12/24/2022



BC210418084: FARIA RAUF

Time Left

88

sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 02:02 PM, 24 December 2022

Question # 10 of 10 (start time: 02:10:14 PM, 24 December 2022)

Total Marks: 1

What is annihilator operator of the function

$$g(x) = 4 \sin x$$

?

Select the correct option

Reload Math Equations

☐	$(D^2 + 4)$
☐	$(D^2 - 4)$
☐	$(D^2 + 1)$
☐	$(D^2 - 1)$

Click to Save Answer & Move to Next Question



BC210418084: FARIA RAUF

Time Left
86
sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 02:02 PM, 24 December 2022

Question # 9 of 10 (Start time: 02:09:25 PM, 24 December 2022)

Total Marks: 1

If the auxiliary equation, $am^2 + bm + c = 0$, has the complex roots, $m_1 = \alpha + i\beta$ and $m_2 = \alpha - i\beta$, where α and $\beta > 0$ are real and $i^2 = -1$. Then the solution of the differential equation, $ay'' + by' + cy = 0$, could be _____.

Select the correct option

Reload Math Equations

- | | |
|-----------------------|---|
| ☐ | $y = c_1 e^{(\alpha+i\beta)x} - c_2 e^{(\alpha-i\beta)x}$ |
| ☐ | $y = c_2 e^{(\alpha-i\beta)x}$ |
| ☐ | $y = c_1 e^{(\alpha+i\beta)x} + c_2 e^{(\alpha-i\beta)x}$ |
| ☐ | $y = c_1 e^{(\alpha+i\beta)x}$ |

Click to Save Answer & Move to Next Question



BC200414970: AMNA

Time Left 86
sec(s)

MTH401 – Differential Equations (Quiz#2)

Quiz Start Time: 02:02 PM, 24 December 2022

Question # 10 of 10 (Start time: 02:09:01 PM, 24 December 2022)

Total Marks: 1

The differential operator D possesses the _____ property.

Select the correct option

☐	nonlinearity
☐	closure
☐	inverse
☐	linearity

Click to Save Answer & Move to Next Question





BC210418084: FARIA RAUF

Time Left

84

sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 02:02 PM, 24 December 2022

Question # 7 of 10 (start time: 02:08:04 PM, 24 December 2022)

Total Marks: 1

For a second order linear ODE, if y_1 and y_2 are its solutions, and further $W(y_1, y_2) = 2$, $W_1 = \frac{e^{-x}}{x}$, then $w'_1 =$ _____.

Select the correct option

Reload Math Equations

☐	$\frac{e^{-x}}{2x}$
☐	$\frac{e^{-x}}{x}$
☐	$2x e^{-x}$
☐	$-\frac{e^{-x}}{2x}$

Click to Save Answer & Move to Next Question



BC210418084: FARIA RAUF

Time Left

88

sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 02:02 PM, 24 December 2022

Question # 8 of 10 (Start time: 02:08:50 PM, 24 December 2022)

Total Marks: 1

For non-homogeneous differential equations, the form of the particular solution, $y_p = y_{p1} + y_{p2} + \dots + y_{pn}$, is a linear combination of all the linearly _____ functions generated by repeated differentiation of the input function $g(x)$.

Select the correct option



Reload Math Equations

☐	dependent
☐	independent

Click to Save Answer & Move to Next Question



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Time Left 86
sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 02:02 PM, 24 December 2022

Question # 9 of 10 (start time: 02:07:52 PM, 24 December 2022)

Total Marks: 1

An annihilator operator L for the function, $f(x) = e^{3x}$, is $L =$ _____.

Select the correct option

Reload Math Equations

☐	$D - 3$
☐	$3e^{3x}$
☐	$D - e^{3x}$
☐	e^{3x}

Click to Save Answer & Move to Next Question





BC210418084: FARIA RAUF

Time Left

89

sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 02:02 PM, 24 December 2022

Question # 6 of 10 (Start time: 02:07:20 PM, 24 December 2022)

Total Marks: 1

If a force acts upon a body, the acceleration is produced in the direction of the force whose magnitude is proportional to the magnitude of force, is known as the _____.

Select the correct option



Newton's first law



Hook's law



Newton's third law



Newton's second law

Click to Save Answer & Move to Next Question



BC200414970: AMNA

Time Left 86
sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 02:02 PM, 24 December 2022

Question # 7 of 10 (Start time: 02:06:49 PM, 24 December 2022)

Total Marks: 1

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = 5x + 7$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

☐	$Ax^3 + Bx^2 + Cx + D$
☐	A
☐	$Ax + B$
☐	$Ax^2 + Bx + C$

Click to Save Answer & Move to Next Question





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Time Left

86

sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 02:02 PM, 24 December 2022

Question # 5 of 10 (Start time: 02:06:18 PM, 24 December 2022)

Total Marks: 1

Consider the non-homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = x e^{3x} \cos 4x$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

- | | |
|-----------------------|---|
| ☐ | $A e^{3x} \cos 4x + B e^{3x} \sin 4x$ |
| ☐ | $A \cos 4x + B \sin 4x$ |
| ☐ | $(Ax + B) e^{3x} \cos 4x + (Cx + D) e^{3x} \sin 4x$ |
| ☐ | $(A_1 x^2 + B_1 x + C_1) \cos 4x + (A_2 x^2 + B_2 x + C_2) \sin 4x$ |

Click to Save Answer & Move to Next Question



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Time Left 86
sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 02:02 PM, 24 December 2022

Question # 6 of 10 (start time: 02:06:01 PM, 24 December 2022)

Total Marks: 1

The input function $g(x)$ can have one of the forms: A constant function k ; A polynomial function; _____.

Select the correct option

Reload Math Equations

☐	(I) An exponential function $e^{\alpha x}$
☐	(IV) All (I), (II) and (III).
☐	(II) The trigonometric functions $\sin \beta x$, $\cos \beta x$
☐	(III) Finite sums and products of these functions

Click to Save Answer & Move to Next Question





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Time Left

87

sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 02:02 PM, 24 December 2022

Question # 4 of 10 (Start time: 02:05:25 PM, 24 December 2022)

Total Marks: 1

Consider the non-homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = (9x - 2)e^{5x}$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

- | | |
|-----------------------|-------------------------------------|
| ☐ | $Ae^{5x} \cos 5x + Be^{5x} \sin 5x$ |
| ☐ | $(Ax + B)e^{5x}$ |
| ☐ | Ae^{5x} |
| ☐ | $(Ax^2 + Bx + C)e^{5x}$ |

Click to Save Answer & Move to Next Question



BC200414970: AMNA

Time Left 87
sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 02:02 PM, 24 December 2022

Question # 5 of 10 (start time: 02:05:14 PM, 24 December 2022)

Total Marks: 1

What is annihilator operator of the function:

$$g(x) = 4 \sin x$$

?

Select the correct option

Reload Math Equations

☐	$(D^2 + 4)$
☐	$(D^2 - 1)$
☐	$(D^2 + 1)$
☐	$(D^2 - 4)$

Click to Save Answer & Move to Next Question





BC210418084: FARIA RAUF

Time Left

88

sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 02:02 PM, 24 December 2022

Question # 3 of 10 (Start time: 02:04:38 PM, 24 December 2022)

Total Marks: 1

If the auxiliary equation $m^2 + m = 0$ has roots $m = 0, m = \pm i$ then the complementary function is

Select the correct option



Reload Math Equations



$$y_c = c_1 \cos x + c_2 \sin x$$



$$y_c = c_1 + c_2 \cos x + c_3 \sin x$$

Click to Save Answer & Move to Next Question



BC200414970: AMNA

Time Left 82
sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 02:02 PM, 24 December 2022

Question # 4 of 10 (start time: 02:04:25 PM, 24 December 2022)

Total Marks: 1

The combination of the Newton's second law and the Hook's law could lead to a differential equation governing to the motion of a mass attached to spring i.e. _____ motion.

Select the correct option

☐	rotational
☐	translational
☐	linear
☐	simple harmonic

Click to Save Answer & Move to Next Question





BC210418084: FARIA RAUF

Time Left 86
sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 02:02 PM, 24 December 2022

Question # 2 of 10 (Start time: 02:03:51 PM, 24 December 2022)

Total Marks: 1

#

$$y = 2 + x$$

, then which of the following is true for it?

Select the correct option

Reload Math Equations

☐	Its annihilator operator is D^2
☐	Its annihilator operator is D
☐	Its annihilator operator is D^3
☐	Its annihilator operator is $D+1$

Click to Save Answer & Move to Next Question



Quiz

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Time Left 85
sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 02:02 PM, 24 December 2022

Question # 3 of 10 (start time: 02:03:42 PM, 24 December 2022)

Total Marks: 1

The equation of free un-damped motion is _____

Select the correct option

Reload Math Equations



$$\frac{d^2x}{dt^2} - \frac{k}{m}x = 0$$



$$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$$

Click to Save Answer & Move to Next Question





BC200414970: AMNA

Time Left 89
sec(s)

MTH401 – Differential Equations (Quiz#2)

Quiz Start Time: 02:02 PM, 24 December 2022

Question # 2 of 10 (Start time: 02:02:54 PM, 24 December 2022)

Total Marks: 1

The polynomial linear differential operators with constant coefficients _____.

Select the correct option

☐	distributive
☐	inverse
☐	identical
☐	commute

Click to Save Answer & Move to Next Question





BC210418240: SHIZA RAUF

Time Left 89
sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 01:54 PM, 24 December 2022

Question # 10 of 10 (start time: 02:01:13 PM, 24 December 2022)

Total Marks: 1

If

$$y = e^{5x}$$

, then which of the following is true for it?

Select the correct option

Reload Math Equations

☐	Its annihilator operator is $(D - 5)^2$
☐	Its annihilator operator is $(D - 5)$
☐	Its annihilator operator is $(D + 5)^2$
☐	Its annihilator operator is $(D + 5)$

Click to Save Answer & Move to Next Question





BC210418084: FARIA RAUF

Time Left 87
sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 02:02 PM, 24 December 2022

Question # 1 of 10 (Start time: 02:02:42 PM, 24 December 2022)

Total Marks: 1

The periodic time is given by

Select the correct option

Reload Math Equations

☐	$\frac{\omega}{2\pi}$
☐	$\frac{2\pi}{\omega}$
☐	$\frac{\pi}{\omega}$
☐	$2\pi + \omega$

Click to Save Answer & Move to Next Question



BC200414970: AMNA

Time Left 87
sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 02:02 PM, 24 December 2022

Question # 1 of 10 (Start time: 02:02:00 PM, 24 December 2022)

Total Marks: 1

We can derive a differential equation governing the motion of a mass attached to spring when the Newton's second law combined with

Select the correct option

☐	Newton's 3rd law
☐	Hook's law

Click to Save Answer & Move to Next Question





BC210418240: SHIZA RAUF

Time Left 89
sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 01:54 PM, 24 December 2022

Question # 9 of 10 (start time: 02:00:41 PM, 24 December 2022)

Total Marks: 1

Wronskian, $W(x, -3x) = \underline{\hspace{2cm}}$.

Select the correct option

Reload Math Equations

☐	1
☐	$3x$
☐	$-3x$
☐	0

Click to Save Answer & Move to Next Question





BC210418240: SHIZA RAUF

Time Left 88
sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 01:54 PM, 24 December 2022

Question # 6 of 10 (start time: 01:58:23 PM, 24 December 2022)

Total Marks: 1

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = \sin 4x$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

☐	$A \cos 4x + B \sin 4x$
☐	$Ax^2 + Bx + C$
☐	Ae^{4x}
☐	$(A + B)(\cos 4x + \sin 4x)$

Click to Save Answer & Move to Next Question





BC210418240: SHIZA RAUF

Time Left 89
sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 01:54 PM, 24 December 2022

Question # 3 of 10 (start time: 01:56:43 PM, 24 December 2022)

Total Marks: 1

For a second order linear ODE, if y_1 and y_2 are its solutions, and further $W(y_1, y_2) = 2$, $W_1 = \frac{e^{-x}}{x}$,
then $u'_1 =$ _____.

Select the correct option

Reload Math Equations

☐	$2x e^{-x}$
☐	$\frac{e^{-x}}{x}$
☐	$-\frac{e^{-x}}{2x}$
☐	$\frac{e^{-x}}{2x}$

Click to Save Answer & Move to Next Question





BC210418240: SHIZA RAUF

Time Left 89
sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 01:54 PM, 24 December 2022

Question # 4 of 10 (start time: 01:57:20 PM, 24 December 2022)

Total Marks: 1

The annihilator operator for the function, $g(x) = 5x$, is _____.

Select the correct option

Reload Math Equations

☐	$D + 1$
☐	D^2
☐	D
☐	$D + 5$

Click to Save Answer & Move to Next Question





BC210418240: SHIZA RAUF

Time Left 89
sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 01:54 PM, 24 December 2022

Question # 7 of 10 (start time: 01:58:54 PM, 24 December 2022)

Total Marks: 1

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = 5x^2 \sin 4x$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

☐	$A \cos 4x + B \sin 4x$
☐	$(A_1 x^2 + B_1 x + C_1) \cos 4x + (A_2 x^2 + B_2 x + C_2) \sin 4x$
☐	$(Ax + B)e^{3x} \cos 4x + (Cx + D)e^{3x} \sin 4x$
☐	$Ae^{3x} \cos 4x + Be^{3x} \sin 4x$

Click to Save Answer & Move to Next Question



BC210418240: SHIZA RAUF

Time Left 89
sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 01:54 PM, 24 December 2022

Question # 8 of 10 (start time: 02:00:07 PM, 24 December 2022)

Total Marks: 1

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = e^{3x} \sin 4x$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

- | | |
|-----------------------|-------------------------------------|
| ☐ | $(Ax + B)e^{5x}$ |
| ☐ | $Ae^{3x} \cos 4x + Be^{3x} \sin 4x$ |
| ☐ | $(Ax^2 + Bx + C)e^{5x}$ |
| ☐ | Ae^{5x} |

Click to Save Answer & Move to Next Question





BC210418240: SHIZA RAUF

Time Left 89
sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 01:54 PM, 24 December 2022

Question # 5 of 10 (start time: 01:57:55 PM, 24 December 2022)

Total Marks: 1

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = 3x^2 - 2$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

☐	$Ax + B$
☐	$Ax^2 + Bx + C$
☐	$Ax^3 + Bx^2 + Cx + D$
☐	A

Click to Save Answer & Move to Next Question





BC210418240: SHIZA RAUF

Time Left 88
sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 01:54 PM, 24 December 2022

Question # 1 of 10 (Start time: 01:54:05 PM, 24 December 2022)

Total Marks: 1

The derivatives $u_1', u_2', \dots, u_n'$ of the unknown functions $u_1, u_2, \dots, u_n$ are find by using

Select the correct option

Reload Math Equations



$$u_n' = \frac{W_k}{W'}, \quad k = 1, 2, \dots, n$$



$$u_n' = \frac{W'}{W_k}, \quad k = 1, 2, \dots, n$$

Click to Save Answer & Move to Next Question



BC210418240: SHIZA RAUF

Time Left 89
sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 01:54 PM, 24 December 2022

Question # 2 of 10 (start time: 01:56:07 PM, 24 December 2022)

Total Marks: 1

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = \text{constant}$ (say 1), then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

☐	$Ax^3 + Bx^2 + Cx + D$
☐	$Ax + B$
☐	A
☐	$Ax^2 + Bx + C$

Click to Save Answer & Move to Next Question



1:43



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BC210418258: HUMAIRA CHANNA

Time Left 86
sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 01:42 PM, 24 December 2022

Question # 3 of 10 (start time: 01:43:55 PM, 24 December 2022)

Total Marks: 1

What is annihilator operator of the function:

$$g(x) = 5$$

?

Select the correct option

Reload Math Equations

☐	$D + D$
☐	$5D$
☐	$D + 1$
☐	D

Click to Save Answer & Move to Next Question



1:42

   67

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BC210418258: HUMAIRA CHANNA

Time Left 84
sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 01:42 PM, 24 December 2022

Question # 1 of 10 (Start time: 01:42:29 PM, 24 December 2022)

Total Marks: 1

An annihilator operator L for the function, $f(x) = e^{3x}$, is $L =$ _____.

Select the correct option

 Reload Math Equations

☐	e^{3x}
☐	$3e^{3x}$
☐	$D - e^{3x}$
☐	$D - 3$

Click to Save Answer & Move to Next Question



1:45



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BC210418258: HUMAIRA CHANNA

Time Left 88
sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 01:42 PM, 24 December 2022

Question # 6 of 10 (start time: 01:45:59 PM, 24 December 2022)

Total Marks: 1

How are frequency and period related in simple harmonic motion?

Select the correct option

- | | |
|-----------------------|----------------------------|
| ☐ | They are directly related |
| ☐ | Their sum is constant |
| ☐ | They are inversely related |
| ☐ | None of the above |

Click to Save Answer & Move to Next Question



1:47



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BC210418258: HUMAIRA CHANNA

Time Left 87
sec(s)

MTH401 – Differential Equations (Quiz#2)

Quiz Start Time: 01:42 PM, 24 December 2022

Question # 9 of 10 (start time: 01:48:09 PM, 24 December 2022)

Total Marks: 1

Vibration of an object on equilibrium point is called simple harmonic motion when the restoring force is proportional to _____

Select the correct option

☐	Time
☐	A spring constant
☐	Displacement
☐	Mass

Click to Save Answer & Move to Next Question



1:46



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BC210418258: HUMAIRA CHANNA

Time Left 90
sec(s)

MTH401 – Differential Equations (Quiz#2)

Quiz Start Time: 01:42 PM, 24 December 2022

Question # 7 of 10 (start time: 01:46:39 PM, 24 December 2022)

Total Marks: 1

The differential operator D transforms a _____ function into another function.

Select the correct option

- | | |
|-----------------------|----------------|
| ☐ | integrable |
| ☐ | discontinuous |
| ☐ | continuous |
| ☐ | differentiable |

Click to Save Answer & Move to Next Question





BC210418258: HUMAIRA CHANNA

Time Left

88

sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 01:42 PM, 24 December 2022

Question # 4 of 10 (start time: 01:44:35 PM, 24 December 2022)

Total Marks: 1

The root(s) of the auxiliary equation, $am^2 + bm + c = 0$, for the differential equation, $ay'' + by' + cy = 0$, is (are) _____.

Select the correct option



Reload Math Equations

- | | |
|-----------------------|---|
| ☐ | (II) $m_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$ |
| ☐ | (IV) Both (I) and (II). |
| ☐ | (I) $m_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$ |
| ☐ | (III) $m_3 = \frac{b + \sqrt{b^2 + 4ac}}{2a}$ |

Click to Save Answer & Move to Next Question





BC210418258: HUMAIRA CHANNA

Time Left

88

sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 01:42 PM, 24 December 2022

Question # 8 of 10 (start time: 01:47:29 PM, 24 December 2022)

Total Marks: 1

The input function $g(x)$ can have one of the forms: A constant function k ; A polynomial function; _____.

Select the correct option

 Reload Math Equations

☐	(IV) All (I), (II) and (III).
☐	(III) Finite sums and products of these functions
☐	(II) The trigonometric functions $\sin \beta x$, $\cos \beta x$
☐	(I) An exponential function $e^{\alpha x}$

Click to Save Answer & Move to Next Question



1:42

   67

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BC210418258: HUMAIRA CHANNA

Time Left

88

sec(s)

MTH401 – Differential Equations (Quiz#2)

Quiz Start Time: 01:42 PM, 24 December 2022

Question # 2 of 10 (start time: 01:43:16 PM, 24 December 2022)

Total Marks: 1

The differential operator L is called an annihilator operator of the function f if:

Select the correct option



(iv) All (i), (ii) and (iii).

(ii) $y = f(x)$ defines a sufficiently differentiable function.(i) L is a linear differential operator with constant coefficients.(iii) The function f is such that $L(y)=0$.

Click to Save Answer & Move to Next Question



1:44



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BC210418258: HUMAIRA CHANNA

Time Left 88
sec(s)

MTH401 – Differential Equations (Quiz#2)

Quiz Start Time: 01:42 PM, 24 December 2022

Question # 5 of 10 (start time: 01:45:18 PM, 24 December 2022)

Total Marks: 1

The polynomial linear differential operators with constant coefficients _____.

Select the correct option

- | | |
|-----------------------|--------------|
| ☐ | inverse |
| ☐ | commute |
| ☐ | distributive |
| ☐ | identical |

Click to Save Answer & Move to Next Question



1:48



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BC210418258: HUMAIRA CHANNA

Time Left 84
sec(s)

MTH401 – Differential Equations (Quiz#2)

Quiz Start Time: 01:42 PM, 24 December 2022

Question # 10 of 10 (Start time: 01:48:47 PM, 24 December 2022)

Total Marks: 1

The differential operator D possesses the _____ property.

Select the correct option

☐	linearity
☐	closure
☐	inverse
☐	nonlinearity

Click to Save Answer & Move to Next Question





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4



BC200405508: MUHAMMAD ADIL LATIF

Time Left 89 sec(s)

MTH401 – Differential Equations (Quiz#2)

Quiz Start Time: 01:33 PM, 24 December 2022

Question # 10 of 10 (start time: 01:40:25 PM, 24 December 2022)

Total Marks: 1

How are frequency and period related in simple harmonic motion?

Select the correct option

- ☐ Their sum is constant
- ☐ They are directly related
- ☐ They are inversely related
- ☐ None of the above

Click to Save Answer & Move to Next Question



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4



BC200405508: MUHAMMAD ADIL LATIF

Time Left 85 sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 01:33 PM, 24 December 2022

Question # 9 of 10 (start time: 01:39:42 PM, 24 December 2022)

Total Marks: 1

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = 5x + 7$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

☐	$Ax^2 + Bx + C$
☐	$Ax + B$
☐	A
☐	$Ax^3 + Bx^2 + Cx + D$

Click to Save Answer & Move to Next Question

What is annihilator operator of the function

$$g(x) = 4 \sin x$$

?

Select the correct option

 Reload Math Equations



$$(D^2 - 4)$$



$$(D^2 - 1)$$



$$(D^2 + 1)$$



$$(D^2 + 4)$$

Click to Save Answer & Move to Next Question



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4



BC200405508: MUHAMMAD ADIL LATIF

Time Left 90 sec(s)

MTH401 – Differential Equations (Quiz#2)

Quiz Start Time: 01:33 PM, 24 December 2022

Question # 8 of 10 (start time: 01:38:53 PM, 24 December 2022)

Total Marks: 1

Vibration of an object on equilibrium point is called simple harmonic motion when the restoring force is proportional to _____.

Select the correct option

- ☐ Displacement
- ☐ Mass
- ☐ A spring constant
- ☐ Time

Click to Save Answer & Move to Next Question



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4



BC200405508: MUHAMMAD ADIL LATIF

Time Left 89 sec(s)

MTH401 – Differential Equations (Quiz#2)

Quiz Start Time: 01:33 PM, 24 December 2022

Question # 5 of 10 (start time: 01:36:40 PM, 24 December 2022)

Total Marks: 1

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = e^{5x}$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

☐	Ae^{5x}
☐	$(Ax + B)e^{5x}$
☐	$(Ax^2 + Bx + C)e^{5x}$
☐	$Ae^{5x} \cos 5x + Be^{5x} \sin 5x$

Click to Save Answer & Move to Next Question



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4



BC200405508: MUHAMMAD ADIL LATIF

Time Left 89 sec(s)

MTH401 – Differential Equations (Quiz#2)

Quiz Start Time: 01:33 PM, 24 December 2022

Question # 7 of 10 (start time: 01:38:14 PM, 24 December 2022)

Total Marks: 1

The differential operator L is called an annihilator operator of the function f if: _____.

Select the correct option

- ☐ (iii) The function f is such that $L(y)=0$.
- ☐ (ii) $y = f(x)$ defines a sufficiently differentiable function.
- ☐ (i) L is a linear differential operator with constant coefficients.
- ☐ (iv) All (i), (ii) and (iii).

Click to Save Answer & Move to Next Question



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4



BC200405508: MUHAMMAD ADIL LATIF

Time Left 89 sec(s)

MTH401 – Differential Equations (Quiz#2)

Quiz Start Time: 01:33 PM, 24 December 2022

Question # 6 of 10 (start time: 01:37:14 PM, 24 December 2022)

Total Marks: 1

If the auxiliary equation has complex roots

$$a \pm i\beta$$

of a differential equation, then the general solution is _____.

Select the correct option

Reload Math Equations

☐	$y = c_1 e^{a+i\beta} + c_2 e^{a-i\beta}$
☐	$y = c_1 e^a + c_2 e^{i\beta}$
☐	$y = c_1 e^{(a+i\beta)x} + c_2 e^{(a-i\beta)x}$
☐	$y = (c_1 e^{a+i\beta} + c_2 e^{a-i\beta})x$

Click to Save Answer & Move to Next Question

Question # 7 of 10 (Start time: 01:33:32 PM, 24 December 2022)

Total Marks: 1

If 3,3 are real roots of a differential equation, then the general solution is_____

Select the correct option

 Reload Math Equations

☐	$y = c_1 e^{3x} - c_2 e^{3x}$
☐	$y = c_1 e^{3x}$
☒	$y = c_1 e^{3x} + c_2 e^{3x}$
☐	$y = (c_1 + c_2 x) e^{3x}$

Click to Save Answer & Move to Next Question



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4



BC200405508: MUHAMMAD ADIL LATIF

Time Left 89 sec(s)

MTH401 – Differential Equations (Quiz#2)

Quiz Start Time: 01:33 PM, 24 December 2022

Question # 4 of 10 (start time: 01:35:53 PM, 24 December 2022)

Total Marks: 1

Wronskian of the functions, $y_1(x) = 1$, $y_2(x) = \sin x$, $y_3(x) = \cos x$, is _____.

Select the correct option

Reload Math Equations

☐	0
☐	1
☐	-1
☐	2

Click to Save Answer & Move to Next Question



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4



BC200405508: MUHAMMAD ADIL LATIF

Time Left 89 sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 01:33 PM, 24 December 2022

Question # 3 of 10 (start time: 01:35:19 PM, 24 December 2022)

Total Marks: 1

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = x^2 e^{3x}$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

☐	$Ae^{3x}\cos 5x + Be^{3x}\sin 5x$
☐	$(Ax + B)e^{5x}$
☐	Ae^{3x}
☐	$(Ax^2 + Bx + C)e^{3x}$

Click to Save Answer & Move to Next Question



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4



BC200405508: MUHAMMAD ADIL LATIF

Time Left 89 sec(s)

MTH401 – Differential Equations (Quiz#2)

Quiz Start Time: 01:33 PM, 24 December 2022

Question # 2 of 10 (start time: 01:34:33 PM, 24 December 2022)

Total Marks: 1

The solution of the auxiliary equation determines the _____ of the differential equation.

Select the correct option

☐	degree
☐	linearity
☐	solutions
☐	order

Click to Save Answer & Move to Next Question

Question # 7 of 10 (Start time: 01:33:32 PM, 24 December 2022)

Total Marks: 1

If 3,3 are real roots of a differential equation, then the general solution is_____

select the correct option

 Reload Math Equations

$$y = c_1 e^{3x} + c_2 e^{3x}$$



$$y = c_1 e^{3x} - c_2 e^{3x}$$



$$y = (c_1 + c_2 x) e^{3x}$$



$$y = c_1 e^{3x}$$

Click to Save Answer & Move to Next Question



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4



BC200405508: MUHAMMAD ADIL LATIF

Time Left 88 sec(s)

MTH401 – Differential Equations (Quiz#2)

Quiz Start Time: 01:33 PM, 24 December 2022

Question # 1 of 10 (Start time: 01:33:46 PM, 24 December 2022)

Total Marks: 1

The derivatives $u_1', u_2', \dots, u_n'$ of the unknown functions $u_1, u_2, \dots, u_n$ are find by using

Select the correct option

Reload Math Equations

- | | |
|-----------------------|---|
| ☐ | $u_n' = \frac{W_k'}{W_k}, \quad k = 1, 2, \dots, n$ |
| ☐ | $u_n' = \frac{W_k}{W_k'}, \quad k = 1, 2, \dots, n$ |

Click to Save Answer & Move to Next Question

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = e^{5x}$, then the assumed particular solution y_p could be _____.

Select the correct option

 Reload Math Equations

☐	$Ae^{5x} \cos 5x + Be^{5x} \sin 5x$
☐	Ae^{5x}
☐	$(Ax + B)e^{5x}$
☐	$(Ax^2 + Bx + C)e^{5x}$

Click to Save Answer & Move to Next Question

The annihilator operator for the function, $g(x) = 3$, is _____.

Select the correct option

 Reload Math Equations

☐	$D + 1$
☐	D
☐	$3D$
☐	$D + 3$

[Click to Save Answer & Move to Next Question](#)

Question # 3 of 10 (start time: 01:27:55 PM, 24 December 2022)

Total Marks: 1

An annihilator operator L for the function, $f(x) = e^{3x}$, is $L =$ _____.

Select the correct option

 Reload Math Equations

☐	$D - e^{3x}$
☐	$D - 3$
☐	$3e^{3x}$
☐	e^{3x}

Click to Save Answer & Move to Next Question

Question # 2 of 10 (Start time: 01:26:44 PM, 24 December 2022)

Total Marks: 1

The polynomial linear differential operators with constant coefficients _____.

Select the correct option

☐	commute
☐	distributive
☐	inverse
☐	identical

[Click to Save Answer & Move to Next Question](#)

sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 01:25 PM, 24 December 2022

Question # 1 of 10 (Start time: 01:25:39 PM, 24 December 2022)

Total Marks: 1

The auxiliary equation of the differential equation, $3y''' + 5y'' + 10y' - 4y = 0$, is _____.

Select the correct option

 Reload Math Equations


$$3m^3 + 5m^2 + 10m$$



$$(3m^3 + 5m^2 + 10m - 4) e^{mx} = 0$$



$$3m^3 + 5m^2 + 10m - 4 = 0$$



$$3m^3 + 5m^2 - 4$$

[Click to go to Question 2](#)



Start Quiz

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MTH401 - Differential Equations (Quiz#2)

Total Questions :10

Please read the following instructions carefully!

1. Quiz will be based upon Multiple Choice Questions (MCQs).
2. You have to attempt the quiz online. You can start attempting the quiz any time within given date(s) of a particular subject by clicking the link for Quiz in VULMS.
3. Each question has a fixed time of 90 seconds. So you have to save your answer before 90 seconds. But due to unstable internet speeds, it is recommended that you should save your answer within 60 seconds. While attempting a question, keep an eye on the remaining time.
4. Attempting quiz is unidirectional. Once you move forward to the next question, you can not go back to the previous one. Therefore before moving to the next question, make sure that you have selected the best option.
5. **DO NOT** press **Back Button** / **Backspace Button** while attempting a question, otherwise you will lose that question.
6. **DO NOT** refresh the page unnecessarily, **specially** when following messages appear
Saving_
Question Timeout: Now loading next question...
7. **Javascript MUST be enabled** in your browser; otherwise you will not be able to attempt the quiz.
8. If for any reason, you lose access to internet (like power failure or disconnection of internet), you will be able to attempt the quiz again from the question next to the last shown question. But remember that you have to complete the quiz before expiry of the deadline.
9. If any student failed to attempt the quiz in given time then no re-take or offline quiz will be held.

Important Announcement

1. It has come to the University's knowledge that some students are using unfair means while attempting their quiz.
2. The University is monitoring & analyzing the quiz attempts and if a student is found involved in using unfair means, i.e., helping tools/software, Web Extension, Add-ons etc., an Unfair Means Case (UMC) will be made against such student(s) including cancellation of their quiz.

Start Quiz

Time Left 00 sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 12:24 PM, 24 December 2022

Question # 2 of 10 (Start time: 12:25:38 PM, 24 December 2022)

Total Marks: 1

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x)$.

If the input function, $g(x) = e^{3x}$, then the assumed particular solution y_p could be _____.

Select the correct option

 Reload Math Equations

☐	$(Ax^2 + Bx + C)e^{3x}$
☐	$Ae^{3x}\cos 5x + Be^{3x}\sin 5x$
☐	Ae^{3x}
☐	$(Ax + B)e^{5x}$

Click to Save Answer & Move to Next Question



MTH401 - Differential Equations (Quiz #2)

Question # 3 of 10 (Start time: 12:26:18 PM, 24 December 2022)

The displacement measured below the equilibrium position are _____.

Select the correct option



negative



positive



Question # 5 of 10 (Start time: 12:27:50 PM, 24 December 2022)

The velocity of a particle moving with simple harmonic motion is ----- at the mean position.

Select the correct option

☐

None of these

☐

Zero

☐

Maximum

☐

Minimum

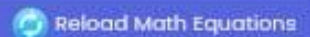
Click to Save Answer & Mark

Question # 7 of 10 (Start time: 12:29:12 PM, 24 December 2022)

Total Marks: 1

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x)$.
 If the input function, $g(x) = x e^{3x} \cos 4x$, then the assumed particular solution y_p could be _____.

Select the correct option



☐	$A e^{3x} \cos 4x + B e^{3x} \sin 4x$
☐	$(A_1 x^2 + B_1 x + C_1) \cos 4x + (A_2 x^2 + B_2 x + C_2) \sin 4x$
☐	$(Ax + B) e^{3x} \cos 4x + (Cx + D) e^{3x} \sin 4x$
☐	$A \cos 4x + B \sin 4x$

[Click to Save Answer & Move to Next Question](#)



MTH401 - Differential Equations (Quiz#2)

Quiz

Question # 6 of 10 (Start time: 12:28:30 PM, 24 December 2022)

What is annihilator operator of the function

$$g(x) = 5$$

?

Select the correct option

☐	D+1
☐	D+5
☐	5D
☐	D

Click to S

Question # 1 of 10 (Start time: 12:24:41 PM, 24 December 2022)

Total Marks: 1

If the auxilliary equation, $am^2 + bm + c = 0$, has two unequal real roots m_1 and m_2 then the solutions of the differential equation, $ay'' + by' + cy = 0$, could be _____.

Select the correct option

 Reload Math Equations

☐

(I) $y = ce^{m_1x}$

☐

(III) $y_2 = e^{m_2x}$

☐

(II) $y_1 = e^{m_1x}$

☐

(IV) Both (II) and (III).

Click to Save Answer & Move to Next Question

Question # 9 of 10 (Start time: 12:31:29 PM, 24 December 2022)

Total Marks: 1

If the auxiliary equation, $am^2 + bm + c = 0$, has the complex roots, $m_1 = \alpha + i\beta$ and $m_2 = \alpha - i\beta$, where α and $\beta > 0$ are real and $i^2 = -1$. Then the solution of the differential equation, $ay'' + by' + cy = 0$, could be _____.

Select the correct option



Reload Math Equations

☐

$$y = c_2 e^{(\alpha - i\beta)x}$$

☐

$$y = c_1 e^{(\alpha + i\beta)x}$$

☐

$$y = c_1 e^{(\alpha + i\beta)x} + c_2 e^{(\alpha - i\beta)x}$$

☐

$$y = c_1 e^{(\alpha + i\beta)x} - c_2 e^{(\alpha - i\beta)x}$$

Click to Save Answer & Move to Next Question



The annihilator operator for the function, $g(x) = 2e^{-x}$, is _____.

Correct option

Reload M

$$D$$

$$D^2$$

$$D + 1$$

$$D + 5$$

Click to Save Answer & Move to Next

Question # 8 of 10 (Start time: 12:30:43 PM, 24 December 2022)

Total Marks: 1

An annihilator operator L for the function, $f(x) = e^{3x}$, is $L =$ _____.

Select the correct option

 Reload Math Equations☐

$$D - e^{3x}$$

☐

$$3e^{3x}$$

☐

$$D - 3$$

☐

$$e^{3x}$$

Click to Save Answer & Move to Next Question

Time Left
sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 12:24 PM, 24 December 2022

Question # 10 of 10 (Start time: 12:32:46 PM, 24 December 2022)

Total Marks: 1

If the auxiliary equation $m^3 + m = 0$ has roots $m = 0, m = \pm i$ then the complementary function is

Select the correct option

 Reload Math Equations

☐ $y_c = c_1 + c_2 \cos x + c_3 \sin x$

☐ $y_c = c_1 \cos x + c_2 \sin x$

Click to Save Answer & Move to Next Question



BC190400727: SHAAF TARIQ

MTH401 – Differential Equations (Quiz#2)

Q1

Question # 10 of 10 (Start time: 10:53:40 AM, 24 December 2022)

The derivatives $u_1', u_2', \dots, u_n'$ of the unknown functions $u_1, u_2, \dots, u_n$ are find by using

Select the correct option

$$u_n' = \frac{W_k}{W}, \quad k = 1, 2, \dots, n$$



$$u_n' = \frac{W}{W_k}, \quad k = 1, 2, \dots, n$$



Click



0727: SHAAF TARIQ

Time Le

Differential Equations (Quiz#2)

Quiz Start Time: 10:45 AM, 24 Decer

9 of 10 (Start time: 10:52:52 AM, 24 December 2022)

Tot

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = \sin 4x$, then the assumed particular solution y_p could be _____.

correct option

Reload Math Editor

$$(A + B)(\cos 4x + \sin 4x)$$

$$A \cos 4x + B \sin 4x$$

$$Ae^{4x}$$

$$Ax^2 + Bx + C$$

Click to Save Answer & Move to Next Qu



Question # 8 of 10 (Start time: 10:52:02 AM, 24 December 2022)

If

$$y = e^{5x}$$

, then which of the following is true for it?

Select the correct option

☐	Its annihilator operator is $(D - 5)^2$
☐	Its annihilator operator is $(D+5)$
☐	Its annihilator operator is $(D-5)$
☐	Its annihilator operator is $(D + 5)^2$

[Click to](#)



Differential Equations (Quiz#2)

Quiz Start Time: 10:45 AM, 24 Dec

10 (Start time: 10:50:58 AM, 24 December 2022)

T

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x)$.

If the input function, $g(x) = 5x^2 \sin 4x$, then the assumed particular solution y_p could be _____.

Select option



Reload Math

$$(A_1 x^2 + B_1 x + C_1) \cos 4x + (A_2 x^2 + B_2 x + C_2) \sin 4x$$

$$A \cos 4x + B \sin 4x$$

$$(Ax + B)e^{3x} \cos 4x + (Cx + D)e^{3x} \sin 4x$$

$$Ae^{3x} \cos 4x + Be^{3x} \sin 4x$$

Click to Save Answer & Move to Next Q



MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 10:45 AM, 24 December

Question # 6 of 10 (start time: 10:49:43 AM, 24 December 2022)

Total Marks

If a force acts upon a body, the acceleration is produced in the direction of the force whose magnitude is proportional to the magnitude of force, is known as the _____.

Select the correct option

- | | |
|-----------------------|---------------------|
| ☐ | Hook's law |
| ☐ | Newton's third law |
| ☐ | Newton's second law |
| ☐ | Newton's first law |

Click to Save Answer & Move to Next Question



Question # 5 of 10 (Start time: 10:49:03 AM, 24 December 2022)

The differential operator D transforms a _____ function into another function.

Select the correct option

☐	integrable
☐	continuous
☐	discontinuous
☐	differentiable



Time L

Differential Equations (Quiz#2)

Quiz Start Time: 10:45 AM, 24 Dec

of 10 (Start time: 10:48:19 AM, 24 December 2022)

To

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x)$.
 If the input function, $g(x) = x^2 e^{5x}$, then the assumed particular solution y_p could be _____.

Correct option

Reload Math

$$Ae^{5x} \cos 5x + Be^{5x} \sin 5x$$

$$(Ax^2 + Bx + C)e^{5x}$$

$$Ae^{5x}$$

$$(Ax + B)e^{5x}$$

Click to Save Answer & Move to Next Q



f 10 (Start time: 10:47:39 AM, 24 December 2022)

The auxiliary equation of the differential equation, $y''' + 2y'' - 3y = 0$, is _____.

ect option



$$m^3 + 2m^2 = 0$$

$$m^3 + 2m^2 - 3 = 0$$

$$(m^3 + 2m^2 - 3) e^{mx} = 0$$

$$m^3 - 3 = 0$$

Click to Save Answer & Move



BC190400727: SHAAF TARIQ

Time Left

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 10:45 AM, 24 Decen

Question # 2 of 10 (Start time: 10:46:47 AM, 24 December 2022)

Tot

The roots of the auxiliary equation, $m^2 - 10m + 25 = (m - 5)^2 = 0$, for a differential equation is _____.

Select the correct option

Reload Math Ec

- | | |
|-----------------------|------------------|
| ☐ | $m_1 = m_2 = 25$ |
| ☐ | $m = -10$ |
| ☐ | $m = 25$ |
| ☐ | $m_1 = m_2 = 5$ |

Click to Save Answer & Move to Next Qu



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Time Left

87
sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 10:45 AM, 24 December 2022

Question # 1 of 10 (Start time: 10:45:56 AM, 24 December 2022)

Total Marks: 1

The combination of the Newton's second law and the Hook's law could lead to a differential equation governing to the motion of a mass attached to spring i.e. _____ motion.

select the correct option



simple harmonic



rotational



linear



translational

Click to Save Answer & Move to Next Question

Question # 10 of 10 (Start time: 10:03:46 AM, 24 December 2022)

Total Marks: 1

If the auxiliary equation, $am^2 + bm + c = 0$, has the complex roots, $m_1 = \alpha + i\beta$ and $m_2 = \alpha - i\beta$, where α and $\beta > 0$ are real and $i^2 = -1$. Then the solution of the differential equation, $ay'' + by' + cy = 0$, could be _____.

Select the correct option

 Reload Math Equations

☐	$y = c_1 e^{(\alpha+i\beta)x}$
☐	$y = c_2 e^{(\alpha-i\beta)x}$
☐	$y = c_1 e^{(\alpha+i\beta)x} - c_2 e^{(\alpha-i\beta)x}$
☐	$y = c_1 e^{(\alpha+i\beta)x} + c_2 e^{(\alpha-i\beta)x}$

Click to Save Answer & Move to Next Question

Question # 9 of 10 (Start time: 10:03:02 AM, 24 December 2022)

Total Marks: 1

The solution of the auxiliary equation determines the _____ of the differential equation.

Select the correct option

☐

degree

☐

order

☐

linearity

☐

solutions

[Click to Save Answer & Move to Next Question](#)

Question # 8 of 10 (start time: 10:02:17 AM, 24 December 2022)

Total Marks: 1

Following is (are) the form(s) of the general solution of the differential equation, $ay'' + by' + cy = 0$, corresponding to the case(s): _____.

Select the correct option

 Reload Math Equations

☐	(I) m_1 and m_2 real and distinct i.e. $(b^2 - 4ac > 0)$
☐	(IV) All (I), (II) and (III).
☐	(III) m_1 and m_2 conjugate complex numbers i.e. $(b^2 - 4ac < 0)$
☐	(II) m_1 and m_2 real and equal i.e. $(b^2 - 4ac = 0)$

[Click to Save Answer & Move to Next Question](#)

Question # 7 of 10 (Start time: 10:01:10 AM, 24 December 2022)

Total Marks: 1

If the auxiliary equation has complex roots

$$\alpha \pm i\beta$$

of a differential equation, then the general solution is _____.

Select the correct option

 Reload Math Equations

☐	$y = c_1 e^{a+i\beta} + c_2 e^{a-i\beta}$
☐	$y = (c_1 e^{a+i\beta} + c_2 e^{a-i\beta})x$
☐	$y = c_1 e^a + c_2 e^{i\beta}$
☐	$y = c_1 e^{(a+i\beta)x} + c_2 e^{(a-i\beta)x}$

Click to Save Answer & Move to Next Question

Question # 6 of 10 (start time: 10:00:08 AM, 24 December 2022)

Total Marks: 1

The roots of the auxiliary equation, $m^2 - 10m + 25 = (m - 5)^2 = 0$, for a differential equation is _____.

Select the correct option



Reload Math Equations

☐	$m_1 = m_2 = 5$
☐	$m = 25$
☐	$m = -10$
☐	$m_1 = m_2 = 25$

Click to Save Answer & Move to Next Question

Question # 5 of 10 (Start time: 09:59:13 AM, 24 December 2022)

Total Marks: 1

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x)$.
 If the input function, $g(x) = e^{3x} \sin 4x$, then the assumed particular solution y_p could be _____.

Select the correct option

 Reload Math Equations

☐	$(Ax^2 + Bx + C)e^{5x}$
☐	Ae^{5x}
☐	$Ae^{3x} \cos 4x + Be^{3x} \sin 4x$
☐	$(Ax + B)e^{5x}$

Click to Save Answer & Move to Next Question

Question # 4 of 10 (Start time: 09:57:44 AM, 24 December 2022)

Total Marks: 1

What is annihilator operator of the function

$$g(x) = 8e^{3x}$$

?

Select the correct option

 Reload Math Equations

☐	$(D - 3)^2$
☐	$(D - 3)$
☐	$(D + 3)$
☐	$(D + 3)^2$

Click to Save Answer & Move to Next Question

Question # 3 of 10 (Start time: 09:56:58 AM, 24 December 2022)

Total Marks: 1

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = 5x + 7$, then the assumed particular solution y_p could be _____.

Select the correct option

 Reload Math Equations

☐	$Ax + B$
☐	$Ax^3 + Bx^2 + Cx + D$
☐	A
☐	$Ax^2 + Bx + C$

[Click to Save Answer & Move to Next Question](#)

Question # 2 of 10 (Start time: 09:54:48 AM, 24 December 2022)

Total Marks: 1

If

$$y = 2 + x$$

, then which of the following is true for it?

Select the correct option

 Reload Math Equations☐Its annihilator operator is D ☐Its annihilator operator is $D+1$ ☐

Its annihilator operator is

$$D^3$$

☐

Its annihilator operator is

$$D^2$$

Click to Save Answer & Move to Next Question

Question # 1 of 10 (Start time: 09:53:48 AM, 24 December 2022)

Total Marks: 1

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x)$.
 If the input function, $g(x) = x^2 e^{5x}$, then the assumed particular solution y_p could be _____.

Select the correct option




$$Ae^{5x}$$



$$(Ax + B)e^{5x}$$



$$(Ax^2 + Bx + C)e^{5x}$$



$$Ae^{5x} \cos 5x + Be^{5x} \sin 5x$$

Click to Save Answer & Move to Next Question

Question # 10 of 10 (Start time: 09:47:36 AM, 24 December 2022)

Total Marks: 1

Wronskian, $W(x, -3x) = \underline{\hspace{2cm}}$.

Select the correct option

 Reload Math Equations

☐	1
☐	0
☐	$3x$
☐	$-3x$

Click to Save Answer & Move to Next Question

Question # 9 of 10 (Start time: 09:46:30 AM, 24 December 2022)

Total Marks: 1

The auxiliary equation of the differential equation, $y''' + 2y'' - 3y = 0$, is _____.

Select the correct option

 Reload Math Equations

$$m^3 - 3 = 0$$



$$(m^3 + 2m^2 - 3) e^{mx} = 0$$



$$m^3 + 2m^2 = 0$$



$$m^3 + 2m^2 - 3 = 0$$

Click to Save Answer & Move to Next Question

Question # 8 of 10 (Start time: 09:45:19 AM, 24 December 2022)

Total Marks: 1

If a force acts upon a body, the acceleration is produced in the direction of the force whose magnitude is proportional to the magnitude of force, is known as the _____.

Select the correct option

☐	Newton's first law
☐	Hook's law
☐	Newton's third law
☐	Newton's second law

[Click to Save Answer & Move to Next Question](#)

Question # 7 of 10 (Start time: 09:44:33 AM, 24 December 2022)

Total Marks: 1

The hook's law states that the force F is proportional to the _____.

Select the correct option

☐

Elongation

☐

Weight

☐

Length

☐

None of these

Click to Save Answer & Move to Next Question

The annihilator operator for the function, $g(x) = 5x$, is _____.

Select the correct option



Reload Math Equations

☐

$$D + 1$$

☐

$$D$$

☐

$$D^2$$

☐

$$D + 5$$

Click to Save Answer & Move to Next Question

The velocity of a particle moving with simple harmonic motion is _____ at the mean position.

select the correct option

☐

Zero

☐

Maximum

☐

None of these

☐

Minimum

Click to Save Answer & Move to Next Question

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = 5x + 7$, then the assumed particular solution y_p could be _____.

Select the correct option

 Reload Math Equations

☐	$Ax^2 + Bx + C$
☐	$Ax^3 + Bx^2 + Cx + D$
☐	A
☐	$Ax + B$

Click to Save Answer & Move to Next Question

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x)$.
 If the input function, $g(x) = (9x - 2)e^{5x}$, then the assumed particular solution y_p could be _____.

Select the correct option

 Reload Math Equations

☐	Ae^{5x}
☐	$(Ax + B)e^{5x}$
☐	$Ae^{5x} \cos 5x + Be^{5x} \sin 5x$
☐	$(Ax^2 + Bx + C)e^{5x}$

Click to Save Answer & Move to Next Question

If

$$y = 2 + x$$

, then which of the following is true for it?

Select the correct option



Reload Math Equations

☐

Its annihilator operator is $D+1$

☐

Its annihilator operator is

$$D^3$$

☐

Its annihilator operator is

$$D^2$$

☐

Its annihilator operator is D

Click to Save Answer & Move to Next Question

Question # 1 of 10 (Start time: 09:37:13 AM, 24 December 2022)

Total Marks: 1

Wronskian of the functions, $y_1(x) = 1$, $y_2(x) = \sin x$, $y_3(x) = \cos x$, is _____.

Select the correct option

 Reload Math Equations

☐	2
☐	-1
☐	0
☐	1

[Click to Save Answer & Move to Next Question](#)